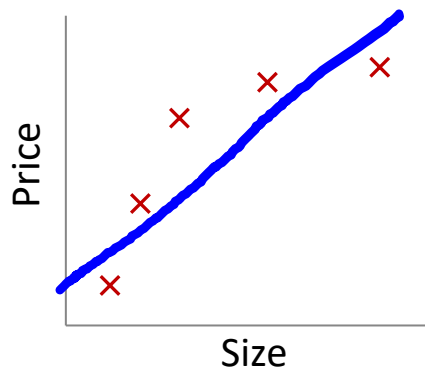


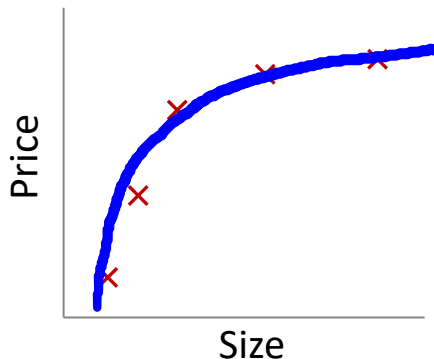
Regularization

The problem of
overfitting

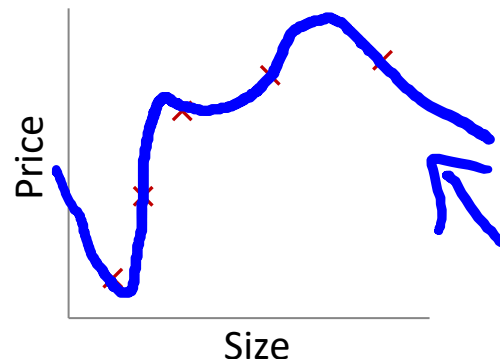
Example: Linear regression (housing prices)



→ $\theta_0 + \theta_1 x$
"Underfit" "High bias"



→ $\theta_0 + \theta_1 x + \theta_2 x^2$
"Just right"



→ $\theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \theta_4 x^4$
"Overfit" "High variance"

Overfitting: If we have too many features, the learned hypothesis may fit the training set very well ($J(\theta) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2 \approx 0$), but fail to generalize to new examples (predict prices on new examples).

Addressing overfitting:

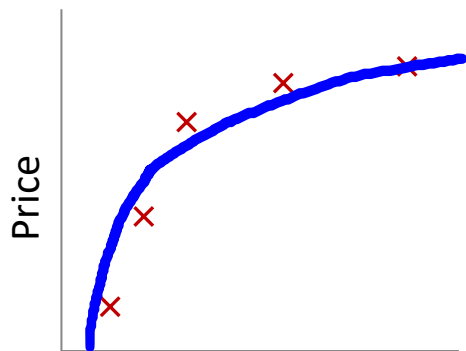
Options:

1. Reduce number of features.
 - — Manually select which features to keep.
 - — Model selection algorithm (later in course).
2. Regularization.
 - — Keep all the features, but reduce magnitude/values of parameters θ_j .
 - Works well when we have a lot of features, each of which contributes a bit to predicting y .

Regularization

Cost function

Intuition



Size of house

$$\theta_0 + \theta_1 x + \theta_2 x^2$$



Size of house

$$\theta_0 + \theta_1 x + \theta_2 x^2 + \cancel{\theta_3 x^3} + \cancel{\theta_4 x^4}$$

↑
↑

Suppose we penalize and make θ_3, θ_4 really small.

$$\rightarrow \min_{\theta} \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2 + \underline{1000 \theta_3^2} + \underline{1000 \theta_4^2}$$

$\underline{\theta_3 \approx 0}$
 $\underline{\theta_4 \approx 0}$

Regularization.

Small values for parameters $\theta_0, \theta_1, \dots, \theta_n$

- “Simpler” hypothesis
- Less prone to overfitting

$$\rightarrow \boxed{\theta_2, \theta_4} \approx 0$$

Housing:

- Features: $x_1, x_2, \dots, x_{100}$
- Parameters: $\theta_0, \theta_1, \theta_2, \dots, \theta_{100}$

$$J(\theta) = \frac{1}{2m} \left[\sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2 + \lambda \sum_{j=1}^n \theta_j^2 \right]$$

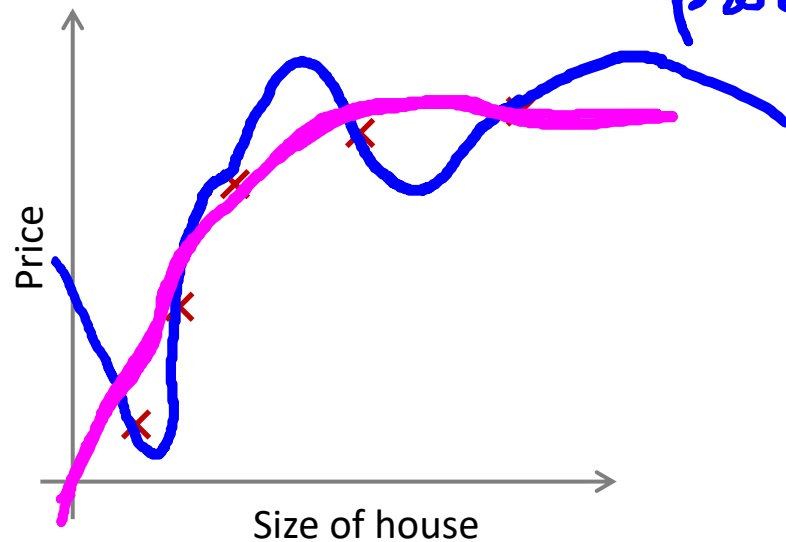
~~$\theta_1, \theta_2, \theta_3, \dots, \theta_{100}$~~

Regularization.

$$\rightarrow J(\theta) = \frac{1}{2m} \left[\sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2 + \lambda \sum_{j=1}^n \theta_j^2 \right]$$

$\min_{\theta} J(\theta)$

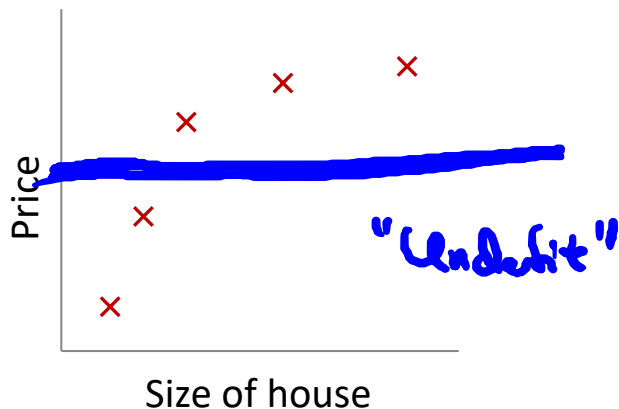
regularization parameter



In regularized linear regression, we choose θ to minimize

$$J(\theta) = \frac{1}{2m} \left[\sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2 + \lambda \sum_{j=1}^n \theta_j^2 \right]$$

What if λ is set to an extremely large value (perhaps far too large for our problem, say $\lambda = 10^{10}$)?



$h_{\theta}(x)$

$$\theta_0 + \cancel{\theta_1 x} + \cancel{\theta_2 x^2} + \cancel{\theta_3 x^3} + \cancel{\theta_4 x^4}$$

$\theta_1, \theta_2, \theta_3, \theta_4$

$\theta_1 \approx 0, \theta_2 \approx 0$

$\theta_3 \approx 0, \theta_4 \approx 0$

$$h_{\theta}(x) = \theta_0$$

Regularization

Regularized linear
regression

Regularized linear regression

$$J(\theta) = \frac{1}{2m} \left[\sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2 + \lambda \sum_{j=1}^n \theta_j^2 \right]$$

$$\min_{\theta} J(\theta)$$



Gradient descent

$$\frac{\partial}{\partial \theta_0}$$

$$\theta_0, \theta_1, \theta_2, \dots, \theta_n$$

Repeat {

$$\frac{\partial}{\partial \theta_0} J(\theta)$$

$$\rightarrow \theta_0 := \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x_0^{(i)}$$

$$\theta_j := \theta_j - \alpha \left[\frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x_j^{(i)} + \frac{\lambda}{m} \theta_j \right] \quad (j = \cancel{0}, 1, 2, 3, \dots, n)$$

$$\theta_j := \theta_j (1 - \alpha \frac{\lambda}{m}) - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x_j^{(i)}$$

$$\rightarrow J(\theta)$$

$$\theta_j^2$$

$$1 - \alpha \frac{\lambda}{m} < 1$$

$$0.99$$

$$\theta_j \times 0.99$$

Normal equation

$$X = \begin{bmatrix} (x^{(1)})^T \\ \vdots \\ (x^{(m)})^T \end{bmatrix}$$

$m \times (n+1)$

$$y = \begin{bmatrix} y^{(1)} \\ \vdots \\ y^{(m)} \end{bmatrix}$$

$\mathbb{R}^m$

$$\min_{\theta} J(\theta)$$

$$\frac{d}{d\theta} J(\theta) \stackrel{!}{=} 0$$

$$\Rightarrow \Theta = (X^T X + \lambda \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{(n+1) \times (n+1)})^{-1} X^T y$$

e.g. $n=2$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Non-invertibility (optional/advanced).

Suppose $m \leq n$, $\leftarrow$
(#examples) (#features)

$$\theta = (X^T X)^{-1} X^T y$$

$\underbrace{(X^T X)^{-1}}_{\text{non-invertible / singular}}$

pinv

inv
 $\mathbb{R}$

If $\lambda > 0$,

$$\theta = \left(X^T X + \lambda \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & \ddots \\ & & & & 1 \end{bmatrix} \right)^{-1} X^T y$$

invertible.

References

- Andrew Ng's slides on Multiple Linear Regression from his Machine Learning Course on Coursera.

Disclaimer

- Content of this presentation is not original and it has been prepared from various sources for teaching purpose.